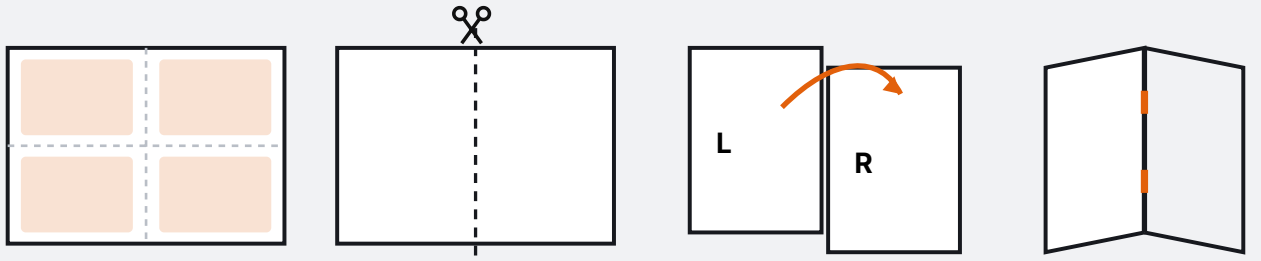
How to print this booklet

This file holds the A6 manual imposed 4 pages per A4 side. Printed double-sided, cut once and folded, it becomes a stapled A6 booklet (56 pages).



1 · Print — 2-sided · short edge · 100 %

2 · Cut

3 · Stack — L1 L2 … / R1 R2 …

4 · Fold and staple

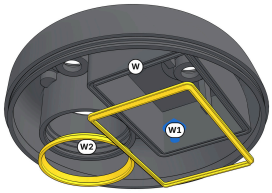
1. Print the whole file double-sided on A4 paper, at actual size (100 %), with “flip on short edge”. This first sheet is the guide: put it aside.
2. Lay the printed sheets with the small labels L1 R1 facing up and in order: sheet 1 on top.
3. Cut the stack in half along the dashed line marked with the scissors.
4. Put the left pile (L1, L2 …) on top of the right pile (R1, R2 …). Do not turn anything over.
5. Fold the whole stack in half along the fold marks, so that L1 stays outside: the cover is now on the front.
6. Staple twice on the fold (a long-arm stapler helps) or sew it. Trim the open edge if you like.

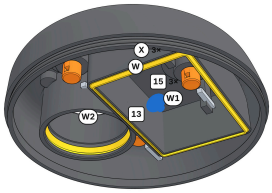
> Try the first two sheets before printing everything: if the backs come out upside down, print again with “flip on long edge”.

No duplex printer? Print the odd pages, put the stack back in the tray, then print the even pages.

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A4 Prepare the cap



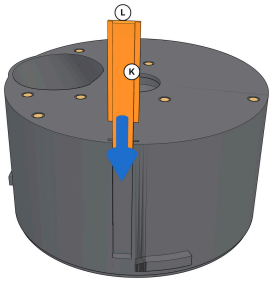


W ×1 W1 ×1 W2 ×1 X ×3 13 ×1 12 ×3 15 ×3

1. Press the cap seal W1 into its D-shaped groove and the chute lip ring W2 into the round socket.
2. Grease the large O-ring 13 and roll it into the groove round the rim.
3. Fit an O-ring 12 on each thumb-turn X, push the thumb-turns up through their holes, then press a key 15 through each shaft.

Check: The 3 thumb-turns turn and cannot fall out.

A5 Turn the housing over and fit the latch

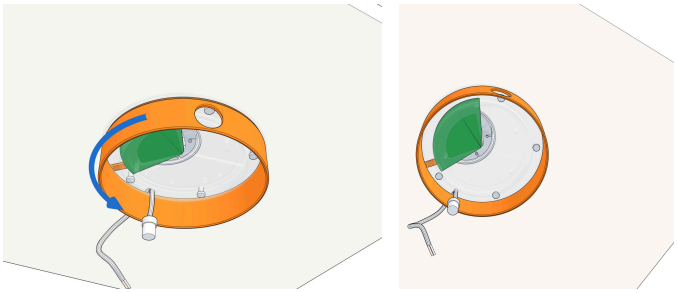


K ×1 L ×1

1. Lay a soft cloth on the table. Put the dosing housing K on it upside down.
2. Slide the safety latch L into its slot in the bottom, small bump outwards, until it is flush.

Check: The latch does not stick out of the bottom.

5 Get to know the gate



1. Turn the orange ring anticlockwise, seen from below, until it stops: the green blade closes the opening. This is CLOSED.
2. In CLOSED the round window of the ring sits where the release button of the unit will be.
3. Turn the ring back until it stops: the opening is free. This is OPEN.
4. Leave the ring in CLOSED for now.

Check: The ring turns smoothly between its two stops (about 1/3 of a turn).

R1 L1

Kibble Dispenser Outdoor

User manual · English

https://ricricucit.github.io/kibble-dispenser/
Scan for the 3D instructions and the design files

0.6.2 draft · design v6.2
Preliminary: not yet printed or tested
Text and pictures CC BY 4.0 · hardware CERN-OHL-P-2.0 · software MIT
© 2026 Enrico Icardi and contributors

EN ENGLISH

Kibble Dispenser Outdoor

User manual

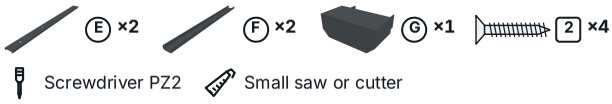
Installation, use and maintenance



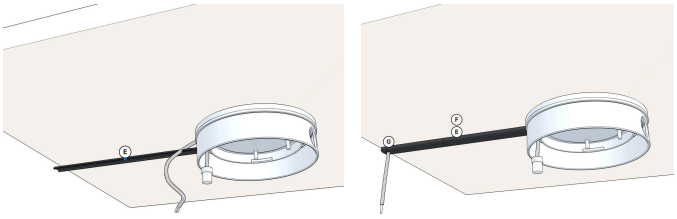
0.6.2 draft design v6.2

⚠ **Preliminary: not yet printed or tested**

6 Fit the cable channel



Screwdriver PZ2 · Small saw or cutter



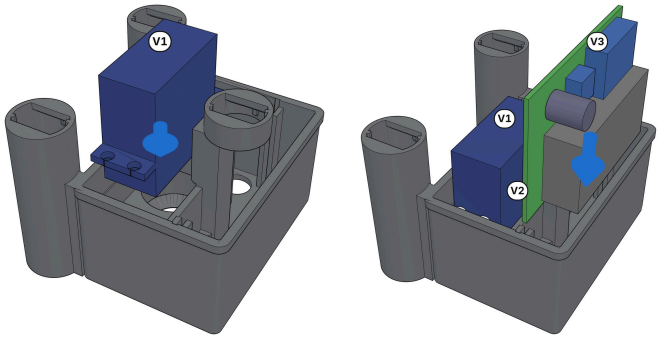
1. Screw the first channel base (E) under the shelf, from the flange towards the back edge, with 2 screws [2].
2. Cut the second base to length so that it ends at the back edge, and screw it on.
3. Lay the cable in the bases, snap the covers (F) on and push the end cap (G) into the end.
4. Lead the cable down through the hole of the end cap.

Check: The covers sit closed along the whole length. No cable is pinched.

A3 Fit out the electronics box



Hex key 2.5 mm, long



1. Set the servo (V1) to its zero position (electronics notes), put it in the cradle and fix it with 4 screws M3 × 8 and washers [8].
2. Push the hub (V2) onto the servo spline with its slot pointing to the gland side, and fix it with 1 screw M3 × 8 [8].
3. Slide the board (V3) up into its slots, parts facing outwards.
4. Screw the cable gland (V4) into the side wall and the membrane vent (V5) into the roof.

Check: The box is clean and dry inside; the short cable with the plug comes out of the gland.

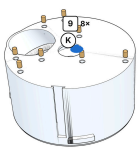
Contents

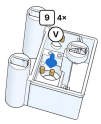
Safety instructions 3
What's in the box 7
Tools you need 9
Installation 10
- Prepare the shelf 10
- Hang and connect 17

First use 21
Everyday use 22
Maintenance 24
Troubleshooting 27
Technical data 29
Spare parts and printable parts 31
Disposal 33
Open source and licences 33
Appendix A · Build it yourself 35
Appendix B · Leak test at home 53
Notes 54

Notes

A1 Fit the threaded inserts





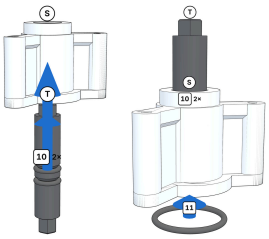
9 ×12

> ⚠ The iron tip is hot: work on a heat-proof surface.

1. Heat each insert 9 with the soldering iron and press it straight into its hole until it is flush.
2. Housing Ⓚ: 8 inserts in the bottom. Electronics box Ⓥ: 4 inserts in the servo cradle.
3. Let them cool before you screw into them.

✓ **Check:** Every insert sits straight and flush with the surface.

A2 Build the drive module



Ⓣ ×1 Ⓢ ×1 10 ×2 11 ×1 16 ×1

1. Grease the 2 O-rings 10 and roll them into the 2 grooves of the shaft Ⓣ.
2. Push the shaft into the tower Ⓢ from below, turning it slightly, until its hexagon sticks out at the top.
3. Grease the large O-ring 11 and press it into the groove under the tower.

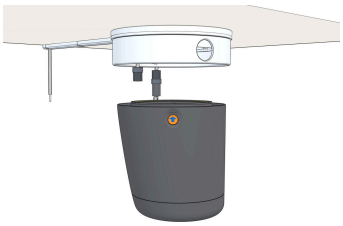
✓ **Check:** The shaft turns by hand with a light, even drag.

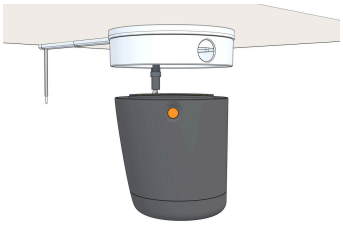
HANG AND CONNECT

7 Connect the unit



Ⓓ ×1





1. Hold the dispenser unit Ⓓ about 8 cm below the flange, release button towards the room. A box or a stool under it helps.
2. Push the plug of the unit into the connector hanging from the flange and screw the coupling ring tight.
3. Push the connector and the loose cable down into the top of the shell.

✓ **Check:** The coupling ring is screwed fully home.

R2 L2

Notes

Safety instructions

Read this manual before you install and use the unit, and keep it. If you pass the unit on, pass the manual on too.

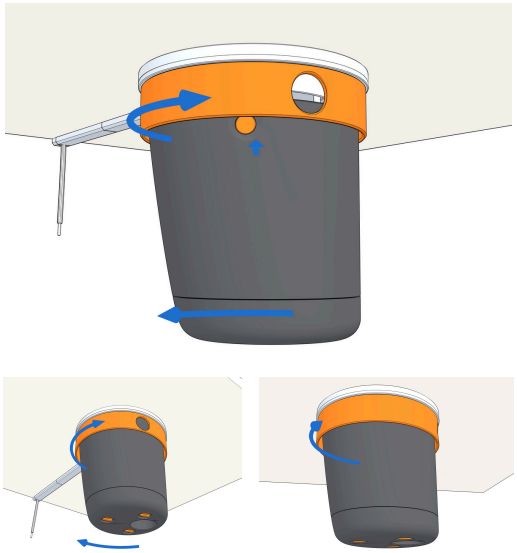
Symbols in this manual

- ⚠ Warning: risk of injury or damage. Read it before you go on.
- ✓ Check: make sure of this before the next step.
- ⓘ Note: useful information.
- Ⓐ Part letter, as listed in What's in the box.
- 1 Screw number, as listed in the hardware bag.

Intended use

- Kibble Dispenser Outdoor gives out portions of dry pet food from a food container that stands on a shelf above it. Each press of a button gives one portion.
- Use it only for dry pet food (kibble) with pieces up to 25 mm. It is not made for wet food, liquids or anything else.
- Install it under a shelf or a table top, indoors, or outdoors under a roof.
- It does not replace your care: check every day that your pet eats and that food comes out.
- This is a preliminary open-source design: no unit has been printed or tested yet. Test it for several days under supervision before you rely on it.

8 Hang the unit



1. Check that the orange ring is in CLOSED.
2. Lift the unit straight up into the flange. Turn it a little until the 3 lugs slip into the slots and it rises.
3. Push up and turn it clockwise, seen from below, by about 1/12 of a turn (30°), until the latch clicks.
4. Turn the orange ring clockwise until it stops: OPEN. Food can now reach the unit.

Check: The unit does not turn back, and the gap under the flange is even all round.

8	**Screw M3 × 8 (+ washer)** servo (4) and hub (1) 1:1	×5
9	**Heat-set insert M3** housing (8) and box (4)	×12
10	**O-ring 9 × 2** shaft	×2
11	**O-ring 24 × 2** tower to box	×1
12	**O-ring 8 × 1.5** thumb-turns	×3
13	**O-ring 132 × 2** cap seam	×1
14	**O-ring 14 × 1.5** release button	×1
15	**Stainless key 3 × 3 × 18** thumb-turns	×3
16	**Food-grade silicone grease (NSF H1)** all O-rings, thinly	×1

Extra tools

- Soldering iron with insert tip
- Hex key 2.5 mm, long
- Soft cloth

Grease every O-ring thinly with food-grade silicone grease before you fit it. The electronics are not final: wire the board as described in the electronics notes.

Electricity

- The unit runs on safe extra-low voltage (12 V DC). Use only the power supply from the kit.
- The power supply is not waterproof: put it in a dry place, protected from rain. Keep the mains plug and the socket dry, and lead the mains cable with a drip loop.
- Do not use the unit if a cable, the power supply or the housing is damaged.
- Unplug the power supply before you take the unit down, open it or clean it.

Moving parts

- Never reach into the outlet, the chute or the shelf hole while the unit is plugged in: the dosing disc turns whenever a button is pressed, and it can move briefly to its rest position when the power comes on.
- Switch the unit off and unplug it before you clear a blockage.

The gate

- Always turn the orange ring to CLOSED before you take the unit down and whenever the food container is removed: otherwise food falls out of the shelf hole.
- Keep a circle of 20 cm free around the unit so that you can reach the ring with your hand.

APPENDIX B

Leak test at home

Optional, for makers: a screening of the seals, not a certification.

1. Take the electronics out of the box, or test before you fit them. Never test with the power connected.
2. Put strips of dry paper towel in the box (at the cable gland, the vent, the 2 screw heads in the roof, along the cap seal and at the thumb-turns) and a few inside the shell, one of them next to the cable grommet under the flange.
3. Close the unit and hang it in its flange set under a shelf or board, as in use, with the gate OPEN.
4. Spray it with a garden hose and a jet nozzle (about 6 mm, about 12.5 L/min: a 10 L bucket fills in 48 s) from 2.5–3 m, from every side and from below, for 3–4 minutes.
5. Dry the outside, take the unit down and open the cap over a towel.

Check: Dry strips in the box: passed. A wet strip shows where water gets in.

For a finer check, weigh a 10 g silica-gel bag on a 0.01 g scale and put it in the box: if it gains less than 0.05 g, no water got in.

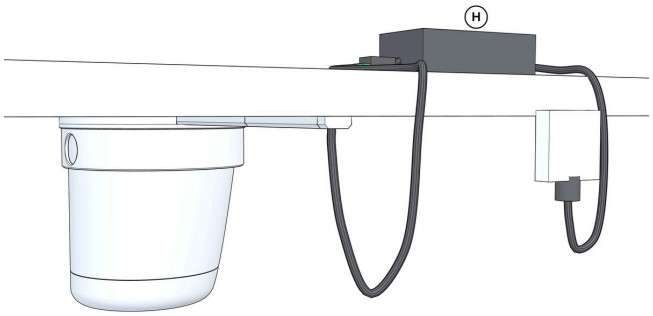
A6	**Cable grommet** TPU 95A yellow	×1

Screws, O-rings and small parts

1	**Wood screw 5 × 35** flange to shelf (5 × 30 for an 18 mm shelf)	×4
2	**Wood screw 3.5 × 16, countersunk** cable channel	×4
3	**Screw M3 × 10** chute	×2
4	**Screw M3 × 12** shell	×3
5	**Screw M3 × 40** box + tower + housing	×3
6	**O-ring 3 × 1** under 2 of the long screw heads	×2
7	**Screw ST2.9 × 9.5, countersunk** joins the two flange plates	×3

9 Connect the power

Ⓗ ×1 Screwdriver PZ2



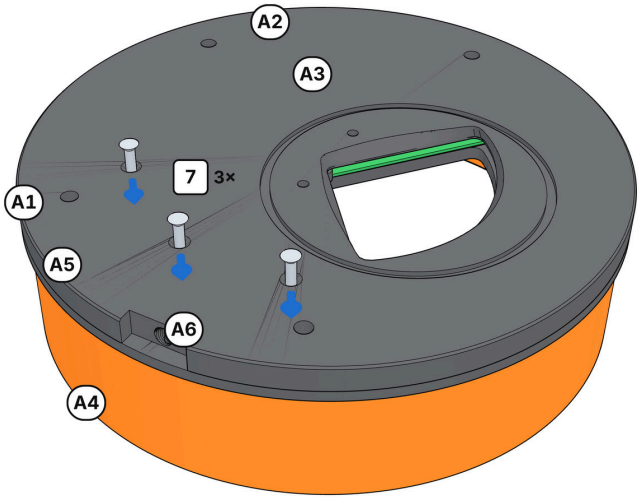
⚠ Mains voltage: the power supply, the plug and the socket must stay dry.

1. Put the power supply Ⓗ in a dry place, protected from rain, for example on the shelf under the roof.
2. Let the cable hang from the end cap in a loop that goes lower than the power supply (drip loop): water drips off at the bottom of the loop.
3. Connect the 2 wires to the adapter: red to +, black to −. Tighten the screws.
4. Plug the power supply into the socket.

✓ **Check:** The unit starts switched OFF, with the LED off. It never gives food when the power comes on.

A16 Build the flange set

CONTINUED



3. Lay the supply cable in the tunnel, press the grommet A6 into the tunnel end on the pod side, then put the upper plate A2 on.
4. Join the plates with 3 screws 7 from the top (the shelf side).

✓ **Check:** Turning the ring moves the blade between OPEN and CLOSED.

ⓘ The unit and the flange set are ready. Now install them like the kit: continue with step 1, page 10.

Children and pets

- This is not a toy. Do not let children play with the unit, its buttons or the packaging.
- Lead the cable and place the power supply where pets cannot chew them.
- Do not let pets climb on the unit, hang from it or play with it.

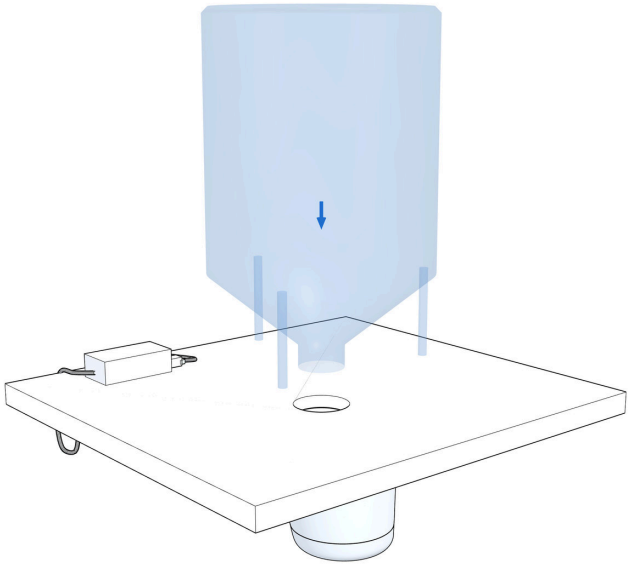
Outdoors

- Outdoors, install the unit only under a roof, out of driving rain and away from sprinklers. Never spray it with a hose or a pressure washer.
- Avoid full sun all day: heat spoils the food faster.
- Seal the joint between the food container and the shelf so that no rain gets into the food.

Food hygiene

- Use only fresh dry pet food in good condition. Throw away food that is damp, mouldy or smells bad.
- Clean the food path every few months and whenever you change the food (page 24).
- The printed parts have fine grooves where crumbs can collect, and they are not certified as food-contact materials.
- Wash the parts by hand only, at 40 °C at most. No dishwasher.

10 Place the container and fill it



⚠ A full container is heavy: make sure the shelf can carry it.

1. Put your food container on the shelf, with its bottom opening right over the green collar.
2. Seal the joint so that no rain gets in.
3. Fill it with dry food, pieces up to 25 mm.

✓ **Check:** With the ring OPEN, food reaches the top of the unit.

During installation

- Wear safety glasses when you drill.
- Check that the shelf can carry the full container: 200 litres of dry food weigh about 80 kg.
- Make sure there are no cables or pipes where you drill.
- Use only the 12 V power supply from the kit. Keep the mains plug and the socket dry.
- Keep small parts away from children and pets.

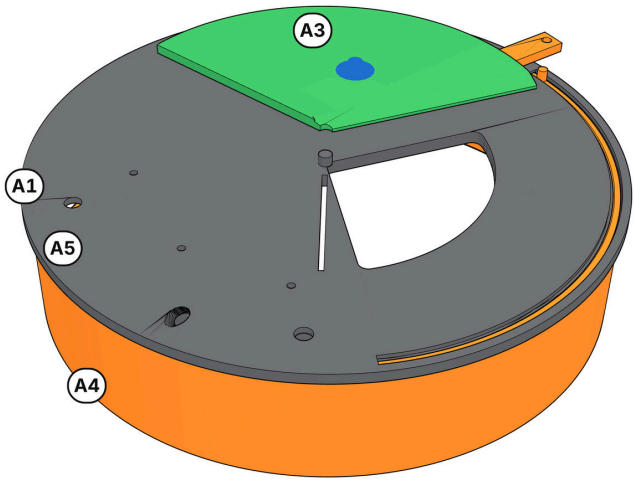
V3	**Circuit board** Bought part	×1
V4	**Cable gland M12** Bought part	×1
V5	**Membrane vent M12** Bought part	×1
W	**Cap** PETG black	×1
W1	**Cap seal** TPU 95A yellow	×1
W2	**Chute lip ring** TPU 95A yellow	×1
X	**Thumb-turns** PETG orange	×3
Y	**Shell** PETG black	×1
Z	**Release button** PETG orange + bought parts	×1
A1	**Lower flange plate** PETG black	×1
A2	**Upper flange plate** PETG black	×1
A3	**Gate blade** PETG green, PETG orange	×1
A4	**Gate ring** PETG orange	×1
A5	**Skirt lip** TPU 95A yellow	×1

A16 Build the flange set

A1 ×1 A2 ×1 A3 ×1

A4 ×1 A5 ×1 A6 ×1 7 ×3

Screwdriver PZ2



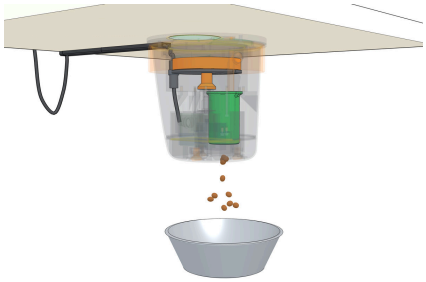
1. Hold the lower flange plate A1 with its 3 hooks pointing down. Press the skirt lip A5 into the groove underneath and snap the orange ring A4 on from below.
2. Put the gate blade A3 on the centre pin from above, with the hole of its orange arm over the peg of the ring.

Printed and bought parts

K	**Dosing housing** PETG black	×1
L	**Safety latch** PETG orange	×1
M	**Rim seal** TPU 95A yellow	×1
N	**Comb lips** TPU 95A yellow	×4
P	**Lower plate** PETG black, PETG green	×1
Q	**Dosing disc** PETG orange, PETG black	×1
R	**Upper plate** PETG black, PETG green	×1
S	**Tower** PETG black	×1
T	**Shaft** PETG black	×1
U	**Chute** PETG green	×1
V	**Electronics box** PETG black + bought parts	×1
V1	**Servo** Bought part	×1
V2	**Hub** PETG black + bought parts	×1

First use

11 First test



1. Put the pet bowl under the outlet.
2. Press ON/OFF: the LED lights, a sound plays and the unit gives one portion.
3. Press the dispense button a few times: one portion and one sound for each press.
4. The first portions can be smaller until the disc fills up.

✓ **Check:** Food falls into the bowl at every press.

THE FIRST DAYS

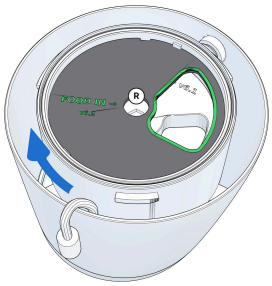
- Check every day that the food reaches the bowl and that your pet eats it.
- Weigh 10 portions on a kitchen scale: then you know how much one press gives with your food.
- Some pets need a few days to get used to the sound.

A15 Fit the upper plate

 Ⓡ ×1

1. Lower the upper plate Ⓡ onto the disc, lips down, FOOD IN on top.
2. Turn it until its 2 notches slide down onto the 2 ribs. It fits only one way.

✓ **Check:** The plate lies flat on its step, just below the rim. The unit is complete.



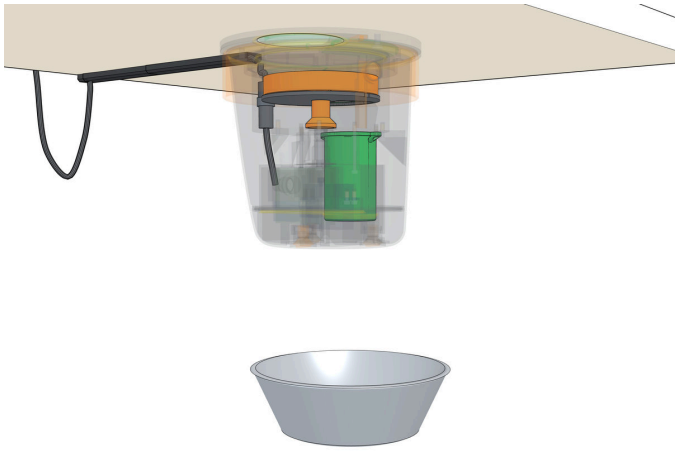
What's in the box

The dispenser unit and the flange set come ready-made: do not take them apart. Check that every part is there.



A	**Flange set** ready-made: two black plates with the green gate blade, the orange ring and the soft yellow skirt	×1
B	**Funnel collar** green ring with 2 small pins	×1
C	**Flange gasket** soft yellow ring	×1

Everyday use



- The ON/OFF button switches the unit on and off. The LED shows when it is on.
- When you switch it on, it gives one portion right away. It never gives food when the power comes on.
- While it is on, each press of the dispense button gives exactly one portion. Holding the button down does not give more.
- A sound plays at every portion. One portion is about 6.5 g with the standard disc.

ⓘ After a power cut the unit starts switched off. The electronics are preliminary: the place of the buttons and the LED is not final yet.

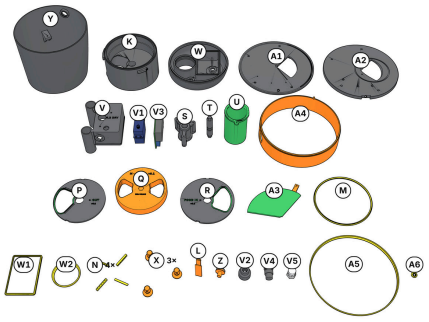
APPENDIX A

Build it yourself

If you print and build the unit and the flange set yourself, do these steps first, then install them like the kit. The online 3D instructions show every step as an animation that you can turn and zoom.

https://ricricucit.github.io/kibble-dispenser/
3D instructions online

Print the parts as described in the print notes of the design. You then build the unit and the flange set yourself.



D	**Dispenser unit** ready-made and closed: the whole pod with the dosing disc, the electronics and a short cable with plug	×1
E	**Channel base** black strip, 2 pieces	×2
F	**Channel cover** black snap-on cover, 2 pieces	×2
G	**Channel end cap** black, with a hole for the cable	×1
H	**Power supply set** 12 V power supply with screw-terminal adapter, supply cable with waterproof connector and yellow grommet	×1
J	**Drilling template** paper, 1:1, A4	×1

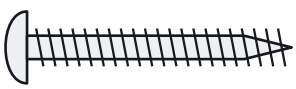
ⓘ You also need a food container (for example a drum) and a level shelf or table top. They are not in the kit.

ⓘ The drive inside is greased for life: you never need grease.

Hardware bag

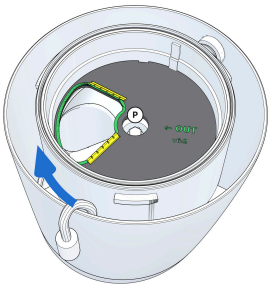
Shown at actual size: hold a screw against the drawing to identify it.

1	**Wood screw 5 × 35** flange to shelf (5 × 30 for an 18 mm shelf)	×4



A13 Fit the lower plate



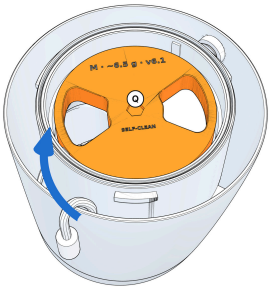


1. Lower the lower plate (P) into the housing, lips up, with the green OUT area over the chute.
2. Turn it until its 2 notches slide down onto the 2 ribs. It fits only one way.

✓ **Check:** The plate lies flat on its step.

A14 Fit the dosing disc





1. Put the dosing disc (Q) on the shaft, label up.
2. Turn it slowly until it drops down. It fits only one way.

✓ **Check:** The disc sits level, with the label on top.

https://ricricucit.github.io/kibble-dispenser/
Project site: 3D instructions, design files, drilling template and updates

Drilling template

The 1:1 drilling template is a separate A4 PDF (kibble-dispenser-drilling-template.pdf, also on the project site). Print it at 100 % (actual size, no scaling) and check the 50 mm scale bar with a ruler. Tape it under the shelf with the PRINTED SIDE FACING YOU and FRONT towards the room.

Open-source hardware is provided as is, without any warranty: see the licence texts.

Using the gate

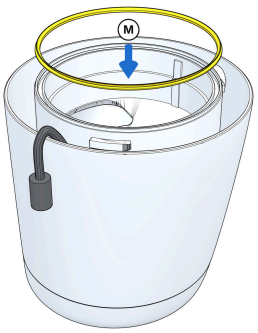


⚠ With the unit taken down, keep the gate CLOSED: otherwise food falls out of the shelf hole.

- OPEN (ring turned clockwise, seen from below): food flows into the unit. This is the normal position.
- CLOSED (ring turned anticlockwise to the stop): the green blade holds the food in the container.
- Close the gate before you take the unit down. Only then does the window of the ring free the release button.
- You can refill the container with the gate open or closed.

⚠ Always turn the ring to CLOSED before you take the unit down or remove the food container.

A11 Stand the unit up and fit the rim seal

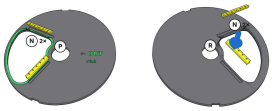

Ⓜ

Ⓜ ×1

1. Turn the unit over: it now stands on its flat bottom. Pull the plug out at the top of the shell and let it hang outside.
2. Press the rim seal Ⓜ into the groove on the top rim of the housing, all the way round. The thin lip points outwards and up.

✓ **Check:** The seal sits evenly in the groove and the lip is not folded.

A12 Press the lips into the plates



Ⓝ ×4 Ⓟ ×1 Ⓡ ×1

1. The 4 comb lips Ⓝ are all the same. Each fits either way round.
2. Lower plate Ⓟ, marked OUT: press 2 lips into the 2 grooves on its top, along the opening.
3. Upper plate Ⓡ, marked FOOD IN: turn it over and press the other 2 lips into the grooves on its underside.
4. The slanted soft flap of each lip leans towards the opening.

✓ **Check:** The lips sit fully in their grooves and stand up evenly.

2 **Wood screw 3.5 × 16, countersunk** ×4
cable channel

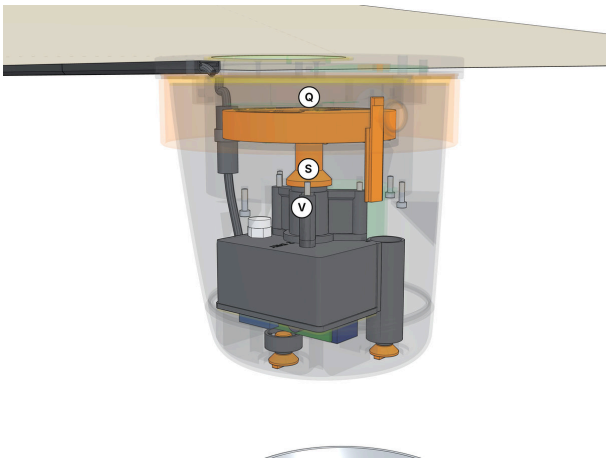


Tools you need

- Drill
- Hole saw Ø 76 mm
- Wood drill bit Ø 3 mm
- Screwdriver PZ2
- Pencil and awl
- Adhesive tape
- Small saw or cutter
- Spirit level
- Safety glasses

Maintenance

Almost no maintenance



- No grease and no oil, ever: the drive inside is sealed and greased for life.
- No silica gel to change: the electronics box breathes through a membrane vent.
- The dosing disc cleans itself: small ribs underneath sweep crumbs away at every turn.
- Keep the 4 tiny drain holes in the flat bottom clear: if one is blocked, open it with a pin.
- For hygiene, clean the food path every few months and whenever you change the food (next two steps).

Disposal

Crossed-out wheeled bin: do not put it in household waste.

The electronics box contains electronic parts. Do not put it in household waste: take it to a collection point for electrical and electronic equipment (WEEE). Dispose of the plastic parts and screws according to local rules. The unit contains no battery.

Printed parts: PETG and TPU. Many household plastic collections do not take them: ask your local recycling centre.

Open source and licences

Kibble Dispenser is open source: design files under CERN-OHL-P-2.0, software under the MIT licence, these instructions under CC BY 4.0. © 2026 Enrico Icardi and contributors.

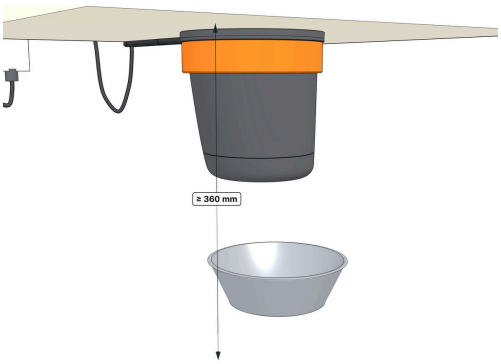
Hardware (design files)	**CERN-OHL-P-2.0**
Software (firmware)	**MIT**
Documentation (this manual, pictures)	**CC BY 4.0**

Installation

PREPARE THE SHELF

1 Choose the place

Spirit level

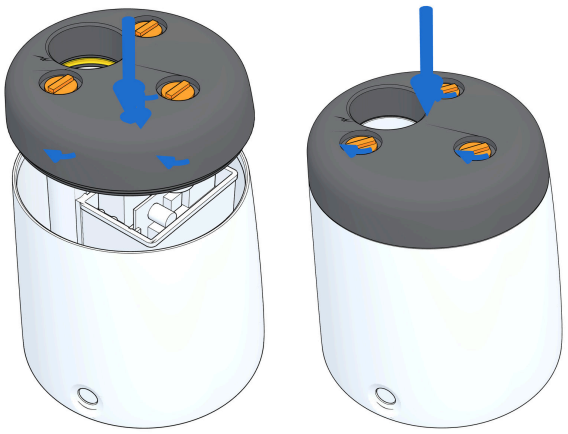


1. Pick a level shelf or table top, 18 to 40 mm thick. Outdoors, choose a place under a roof.
2. Leave at least 36 cm free below the shelf: 18.2 cm for the unit, 8 cm to take it down, and room for the pet bowl.
3. Keep a circle of 20 cm free around the unit: you turn the orange ring by hand.
4. The release button faces the room. The cable runs under the shelf to its back edge.

Check: The shelf is level in both directions.

A10 Close the box with the cap

 Ⓦ ×1



1. Check that the grip ridges of the 3 thumb-turns lie crosswise (open).
2. Put the cap Ⓦ on the box and the shell: the chute goes into the round socket of the cap. It fits only one way.
3. Press the cap firmly home with your palm, with about 13 kg: the black seam O-ring needs it.
4. Keep pressing and turn each thumb-turn a quarter turn clockwise. Release: they click.

Check: The seam is closed all round. The 3 grip ridges are parallel and point towards the round outlet.

SCREWS

1 **Wood screw 5 × 35** ×4
flange to shelf (5 × 30 for an 18 mm shelf)



2 **Wood screw 3.5 × 16, countersunk** ×4
cable channel



ⓘ Print the seals and lips in TPU 95A and all other printed parts in PETG, in the colours shown. The electronics box and its cap must be printed solid (100 % infill) to stay watertight; the print settings are in the design files.

12 Take the unit down



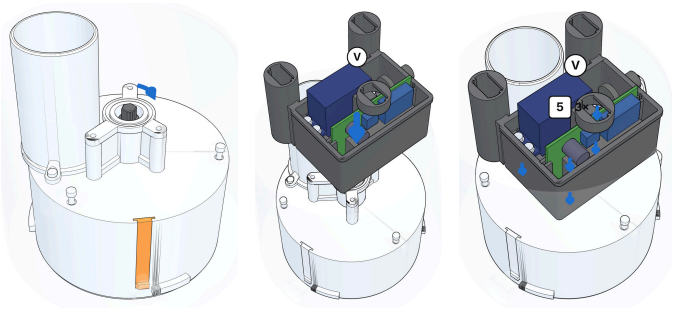
1. Switch the unit off and unplug the power supply.
2. Turn the orange ring anticlockwise, seen from below, to the stop: CLOSED. The window now shows the release button.
3. Press the button through the window and turn the unit anti-clockwise by 30°.
4. Lower it about 8 cm, unscrew the connector in the gap and take the unit down in one piece.

ⓘ The ring can only free the button when the gate is closed, so no food falls out.

A9 Fit the electronics box

 Ⓥ ×1  5 ×3 6 ×2

Hex key 2.5 mm, long

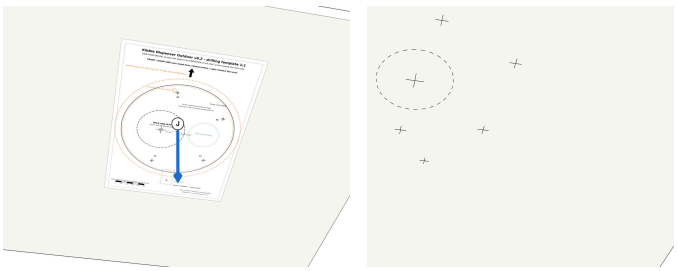


1. Turn the end of the shaft until its small rib points to the orange latch.
2. Lower the box Ⓥ, open side up, into the shell onto the drive module: the shaft goes into the hub. It fits only one way.
3. Put an O-ring 6 under the heads of the 2 screws that sit inside the box.
4. Screw in the 3 long screws M3 × 40 5 with the long hex key. Tighten until snug.
5. Lay the short cable along the housing, so that its plug sits just under the rim of the shell.

✓ **Check:** The box sits flat on the drive module, without a gap.

2 Mark the holes with the template

 Ⓙ ×1 Pencil and awl Adhesive tape

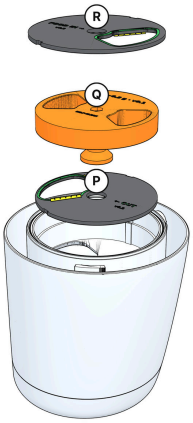


1. Print the drilling template Ⓙ at 100 % and check the 50 mm scale bar.
2. Tape it to the underside of the shelf with the printed side facing you and FRONT towards the room.
3. Prick through the paper with an awl: the centre of the big hole, the 4 screw points and the first channel screw.
4. Take the template off.

✓ **Check:** You have the centre of the big hole, 4 screw points and one channel point.

ⓘ The template is a separate A4 sheet: kibble-dispenser-drilling-template.pdf. Print it at 100 %, PRINTED SIDE FACING YOU under the shelf.

13 Clean the food path



⚠ No dishwasher, no oil, no grease.

1. Stand the unit on the table. Lift out the upper plate, the disc and the lower plate.
2. Wash them and the inside of the housing by hand with warm water (max. 40 °C) and a little dish soap. Leave the lips in place.
3. Let everything dry completely.
4. Refit the lower plate, the disc and the upper plate, each turned until it drops into place. Then connect and hang the unit again (steps 7 and 8).

✓ **Check:** Everything is dry before you refit it.

fold

Spare parts and printable parts

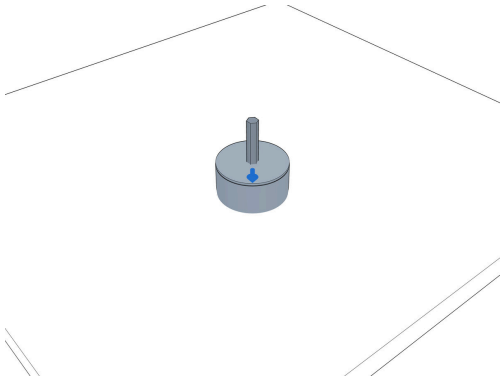
Every printed part can be printed again from the open design files on the project site. Screws, O-rings, the electronics and the power supply are standard bought parts: the parts list in the design files gives their exact type.

A	**Flange set** Ready-made module: replace it as a whole	×1
B	**Funnel collar** PETG green	×1
C	**Flange gasket** TPU 95A yellow	×1
D	**Dispenser unit** Ready-made module: replace it as a whole	×1
E	**Channel base** PETG black	×2
F	**Channel cover** PETG black	×2
G	**Channel end cap** PETG black	×1
H	**Power supply set** Bought part	×1
J	**Drilling template** PDF: print it at 100 %	×1

ⓘ The printable parts of the unit and the flange set, with their colours, are listed in the appendix (page 35).

3 Drill the holes

Drill · Hole saw Ø 76 mm · Wood drill bit Ø 3 mm · Safety glasses



⚠ Wear safety glasses and hold the drill with both hands: the hole saw can jerk.

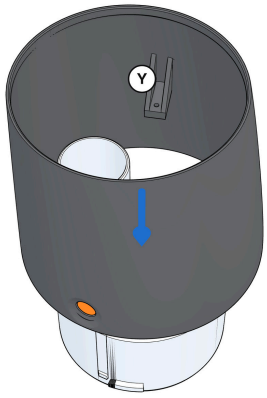
1. Drill straight through at the centre mark with the centre drill of the hole saw.
2. Saw the Ø 76 mm hole from above, guided by the pilot hole. Keep the drill upright.
3. Pre-drill the 4 screw points and the channel point with the Ø 3 mm bit, about 20 mm deep.
4. Remove the chips and smooth the edges.

✓ **Check:** The big hole goes straight through the shelf.

fold

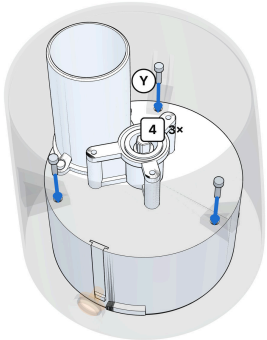
A8 Fit the shell

Ⓨ ×1 · Ⓩ ×1 · 14 ×1 · 4 ×3



1. Grease the small O-ring 14, fit it in the groove of the button hole, and push the release button Ⓩ in from the inside.
2. Lower the shell Ⓨ over the chute and the drive module, with the orange button right next to the orange latch.
3. Fix it with 3 screws M3 × 12 4 through its 3 tabs, reaching in with the long hex key.

✓ **Check:** The shell sits evenly all round.



Gate	in the flange, turned by the orange ring (113°); the unit can only be released with the gate closed
Power	12 V DC, power supply included; M8 IP67 connector; cable in a channel under the shelf
Weather protection	electronics box designed for IP66, inside a sealed outer pod with an insect-proof drain (not tested yet)
Strength	calculated for 250 N down and 100 N sideways at the bottom (not tested yet)
Materials	PETG, TPU 95A seals, NBR O-rings, stainless steel screws and keys, brass threads
Maintenance	no grease, no silica gel

All values are design values: nothing has been printed or tested yet.

Troubleshooting

PROBLEM CAUSE · REMEDY

1 No food comes out

CAUSE	The gate is closed, the unit is off, the container is empty, or food is stuck above the collar.
REMEDY	Turn the ring to OPEN, switch on (LED), refill, tap the container.

2 The disc stops, or you hear grinding

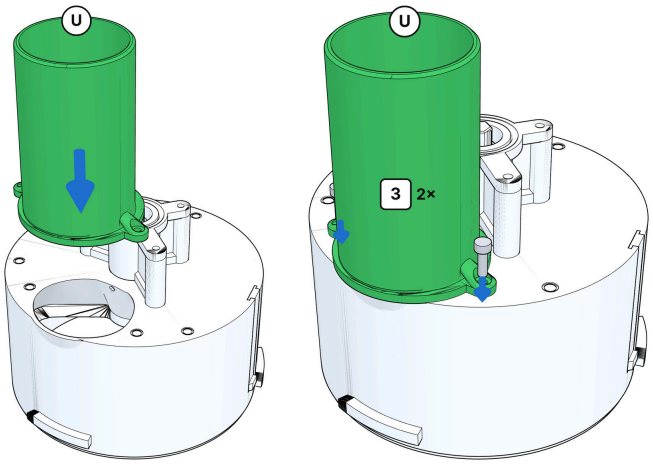
CAUSE	A piece bigger than 25 mm or a hard piece is stuck at an edge.
REMEDY	Switch off, take the unit down (step 12) and clean it (step 13).

3 Portions too small or too big

CAUSE	The portion depends on the food: about 5.4 to 7.4 g from light to heavy kibble.
REMEDY	Press more or fewer times.

A7 Fit the chute

Ⓤ ×1 · 3 ×2 · Hex key 2.5 mm, long

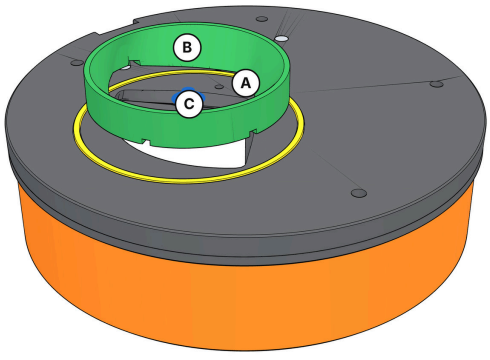


1. Put the green chute Ⓤ on the funnel outlet next to the drive module. It fits only one way.
2. Fix it with 2 screws M3 × 10 3 and the long hex key. Tighten until snug.

Check: The chute stands straight and sits tight against the housing.

4 Fit the flange set

Ⓐ ×1 · Ⓑ ×1 · Ⓒ ×1 · Ⓗ ×1 · 1 ×4 · Screwdriver PZ2



⚠ Shelf thinner than 20 mm: use 4 screws 5 × 30 instead, so the tips do not come through.

1. Press the flange gasket Ⓒ into the groove on the top of the flange set Ⓐ.
2. Put the green collar Ⓑ into the big opening: its 2 pins go into the 2 small holes.
3. Push the bare end of the supply cable from the power supply set Ⓗ into the small hole inside the flange until it comes out at the rim. Press the yellow grommet into the hole: the connector hangs below.

4 Water inside

CAUSE	A leaking container joint, a blocked drain hole or a damaged seal.
REMEDY	Seal the container joint and clear the 4 drain holes. Water in the electronics box: unplug, take the unit down, open the cap (press it and turn the 3 thumb-turns back), let it dry and check the seals.

5 Noise

CAUSE	A soft whirr at each portion is normal. Squeaking or scraping: crumbs between the disc and a plate.
REMEDY	Clean the food path. Never oil anything.

6 The unit does not lock

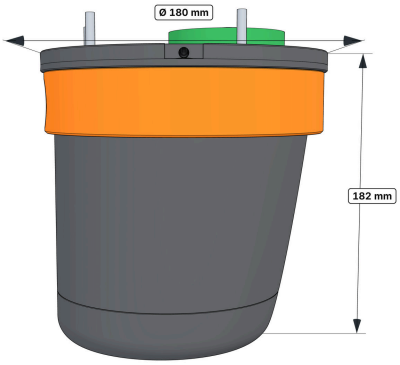
CAUSE	The rim seal is out of its groove, or the lugs are not in the slots.
REMEDY	Seat the rim seal, find the slots, push up firmly while you turn.

7 The release button cannot be reached

CAUSE	The gate is not fully closed: the ring covers the button.
REMEDY	Turn the ring anticlockwise, seen from below, to the stop.

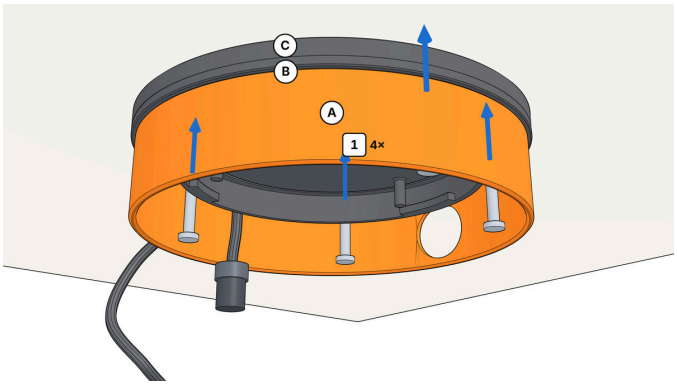
Still not working? Unplug the unit and look for help on the project site (page 33).

Technical data



Design	Kibble Dispenser Outdoor v6.2
Size under the shelf	Ø 180 mm (flange), 182 mm high
Weight of the unit	about 1.3 kg
Shelf	level, 18–40 mm thick; one hole Ø 76 mm
Free space	at least 36 cm below the shelf, Ø 20 cm round the unit
Portion	about 6.5 g (standard disc); 5.4–7.4 g depending on the food
Food size	dry food up to 25 mm

4 Fit the flange set
CONTINUED



4. Lift the flange against the shelf, FRONT mark towards the room: the collar goes into the big hole.
5. Fix it with 4 wood screws [1] and the PZ2 screwdriver. Tighten until snug.

Check: The flange sits tight against the shelf. The collar does not stick out above the shelf.

A6 Set the drive module on the housing



1. Set the drive module on the centre of the housing: the collar goes into the hole, the shaft into the housing.
2. Turn it until its 3 pillars sit over the 3 inserts. It fits only one way. No screws yet.

Check: The module sits flat in the centre.

